

Work on work

Goal. Use the integral, our infinite adding machine, to find the work required to do the following.

1. An underground hemispherical dome with base a circle of radius 20 m is filled with mud of density 1906 kg/m^3 . (Note the dome is oriented so the flat part is farthest below ground.) Find the work done against gravity to pump enough mud out of the dome so that the level of the mud in the dome is 2 m after pumping. The dome is buried 1 meter below ground and the mud travels up the pipe and is dispensed on top of the ground. Draw a relevant picture in the coordinate axes and explain each step of your work.
 - (a) First draw a picture of the dome in the coordinate axes. Draw a horizontal layer at height y .
 - (b) Next, find the area $A(y)$ of the layer at height y as a function of y .
 - (c) Write down an expression for the volume of this layer if it has thickness Δy .
 - (d) What is the mass of this layer?
 - (e) What is the force due to gravity on this layer?
 - (f) What is the vertical distance that this layer is lifted?
 - (g) Use your answers to (e) and (f) to find the work needed to pump the mud in this layer out of the spout.
 - (h) Integrate this expression to solve the problem.

2. A 35 meter rope whose mass is 105 kg is hanging from the top of a building 35 meters high. A 250 kg piano is sitting on the ground attached to the lower end of the rope. How much work is needed to reel in the rope and the piano so that the piano is at a level that is 10 meters below the top of the building? Draw a relevant picture in the coordinate axes and explain each step of your work.
3. A tank of water is 15 feet long and has a cross section in the shape of an equilateral triangle with sides 2 feet long (point of the triangle points directly down). The tank is filled with water to a depth of 9 inches. Determine the amount of work needed to pump all of the water to the top of the tank. Assume that the weight of the water is 62 lb/ft^3 .